ECON 8260
Professor Meghan Skira
Replication and Extension Assignment
Due: Friday, March 20, 2026 by 5pm (EST)

Overview

- Read "Giving Mom a Break: The Impact of Higher EITC Payments on Maternal Health" by W. Evans and C. Garthwaite (2014) in *American Economic Journal: Economic Policy*, 6(2): 258–90.

- You will replicate the statistics, estimation results, and figures that use the BRFSS and NHANES data and you will extend some of that analysis. You may discuss this assignment with your classmates, but everyone must write their own code and answers.

- There is a Stata replication package available here. To access it, create an ICPSR account. Use the final analysis datasets provided in the replication package. You do *not* need to build the analysis datasets from the raw data.

- You may use the code in the replication package. However, that code often does not align with what is described in the paper. For the BRFSS results, view the results in the paper as correct and revise the code to produce those results. For the NHANES results, there are some pre-expansion means for the treatment group that you will not be able to match in Table 7 as well as a handful of point estimates, standard errors, and p-values that you will not be able to match exactly in Tables 6 and 7.

- In line 9 of BRFSS_CODE_REPLICATION.do, you should use BRFSS_Final_Data.dta (provided in the replication package) rather than brfss_data_all_2.dta (not provided in the replication package). This dataset will yield the correct results.

Deliverables

- You will turn in 3 items:

  1. A set of *very clearly-documented* codes (e.g., Stata do files or R scripts).
  2. A set of output files that correspond to the execution of the above codes (e.g., Stata log files or R Markdown document).
  3. A .pdf file that contains the output of the replication and extension exercises described below.

  Write your codes such that I can simply change the directory and execute them. I organize my project folders with the following subfolder structure:

- Data
- Scripts
- Logs
- Output
  - Tables
  - Figures

If you write multiple codes or scripts (e.g., one for BRFSS analyses, one for NHANES analyses), please provide me with a master code that executes all the scripts.[1] You may use any statistical package you like (e.g., R, Stata), but be sure to provide files that are equivalent to the ones described above.

- Your .pdf file will contain a Replication section and an Extension section.
- Your submitted codes should contain *all* the code needed to produce the replicated results as well as the results for the extension. You will need to substantially edit the code provided in the replication package in order to replicate the results in the paper. As mentioned above, the code contains errors and omissions (e.g., code for summary statistics, figures, results described in footnotes). I strongly encourage you to consolidate and streamline the code. For example, many lines of repetitive code can (and should) be condensed using loops and macros, which is better practice than having many repeated lines of code.[2]
- While not required, I *strongly* encourage you to use commands that allow you to output results directly into LaTeX tables rather than cutting and pasting the results into LaTeX(e.g., *eststo* and *esttab* in Stata). Cutting and pasting is considered bad practice and often leads to mistakes. In fact, this paper has mistakes that are likely attributed to cutting and pasting.

Replication Tasks

1. Replicate Tables 2-7. For Table 5, you only need to reproduce the "Observations," "Sample mean," and "Percent with risky levels" columns.
2. Replicate Figure 4. Insert a vertical line at year=1996 as this makes it easier to see pre- versus post-expansion changes in the outcomes. Include an additional subfigure for the "at work" outcome as that is the outcome that is used in the regressions, not "in labor force." Thus, your Figure 4 should contain 5 subfigures.

[1]See the summary.do file in the replication package associated with Garthwaite, Gross, and Notowidigdo (2014; *QJE*) for a good example of a master Stata do file.

[2]There are many resources on good coding practices for economists. Here, here, and here are some I find useful.

3. Create a table titled "Additional Robustness Checks" with the same format as Table 4. Include three sets of results using the BRFSS data. For each set of results, include the corresponding regression-adjusted difference-in-differences coefficient estimates, standard errors, and p-values for all four health outcomes.
    - Estimate the specifications described in footnote 12 and include the results in the first column of this new table.
    - Estimate the specifications described in footnote 21 and include the results in the second column of this new table. To obtain the estimates described in footnote 21, you must exclude a different set of years than what is described in the footnote text.
    - Estimate the specifications as *actually* described in the text of footnote 21 and include the results in the third column of this new table.

In the Replication section of your .pdf document, provide and clearly label your tables and figure and make sure they are legible when printed in black and white.

Extension Tasks

1. Estimate an event-study model based on the regression-adjusted difference-in-differences specification (equation 1 in the paper). That is, using the BRFSS data, estimate:

$$y_{imt} = \alpha + Two_i\phi + \sum_{t=1993}^{2000} T(t)\pi_t + X_i\gamma + \sum_{m=1}^{50} S(m)\lambda_m + \sum_{t=1993}^{2000} (T(t) \times Two_i)\delta_t + \varepsilon_{imt} \quad (1)$$

   The δ_t parameters are the event-study coefficients and measure the evolution of outcomes among those with two or more children in the years before and after the EITC expansion relative to women with one child. Normalize $\delta_{1995} = 0$. In other words, the indicator $T(1995)$ is omitted. Thus, the δ_t coefficients describe the evolution of outcomes relative to 1995, before the EITC extension. All other variables are as described in the paper. Estimate the event-study model for the four main health outcomes using the sample corresponding to the last column of Table 3.

   Plot the δ_t coefficients along with 95 percent confidence intervals for the four main health outcomes (i.e., produce 4 event study figures). Hint: *coefplot* will be helpful if using Stata. The horizontal axis should contain years and the vertical axis represents the event-study coefficient estimates. Note, for number of bad mental health days and number of bad physical health days, estimate linear models via OLS. Do not estimate negative binomial models for the event studies.[3]

[3] Chapter 9.4 of *Causal Inference: The Mixtape* (available here) has some nice examples of how to format and display event-study figures.

2. Given the event-study results, discuss and evaluate the degree to which they support the parallel trends assumption, which is the main identifying assumption in difference-in-differences estimation. Hint: Consider using the HonestDiD package in Stata or R.

In the Extension section of your .pdf document, provide and clearly label your event-study figures and include the discussion of whether they support the parallel trends assumption.

Bonus Question

Studies have recently highlighted concerns about including covariates in difference-in-differences models. Callaway (2023, *Handbook of Labor, Human Resources and Population Economics*) at the end of Section 4.1 as well as Roth et al. (2023, *Journal of Econometrics*) on page 2232 briefly discuss these concerns and cite additional studies on the topic. To what extent do the difference-in-differences models in Evans and Garthwaite (2014) suffer from these concerns?